



UNIVERSITY OF TEHRAN

Electrical and Computer Engineering Department

Digital Logic Design, ECE 367 / Digital Systems I, ECE 894

Spring 1404

Computer Assignment 3

Logic block synthesis, Latches and flip flops, Clocked feedback

Week 11

Name:

Date:

1. Generate a logic block with $A[7:0]$ and $B[7:0]$ inputs and $W[7:0]$ outputs. The logic block has four modes of operation determined by $m[1:0]$ mode inputs. In addition, there is a C_i (carry-in) and a C_o (Carry-out) output.

 $m=00$: W becomes A .
 $m=01$: W becomes $A+1$ if C_i is 1.
 $m=10$: W becomes $A-1$ if C_i is 0.
 $m=11$: W becomes B .
2. Use Yosys and a library of basic logic gates with delay values to synthesize the logic block of Part 1. Generate a Verilog netlist representing the synthesized output. Simulate the pre-synthesis description of Part 1 and the post-synthesis description of this part in a testbench and compare the results. Use advanced testbench constructs like \$random, repeat and other flow control statements.
3. Generate a clocked SR-latch with active high S, R, and Clock inputs. Show this circuit in terms of an all-NAND circuit. Describe this circuit in Verilog using NAND primitives. Annotate the timing of this circuit with gate delays that are based on switch level delays of 4 NS for the NMOS and 6 NS for the PMOS transistors. Simulate this circuit to verify its operation. Apply simultaneous active inputs (S and R) and see the loss of memory.
4. Use an inverter to convert the SR latch of Part 3 to a clocked D-latch. Simulate this circuit to verify its operation. Generate a resetting method to enable it to start with an initial value.
5. Construct an 8-bit latch-register using the above D-latch. Simulate this circuit in a testbench and see its operation.
6. Take the circuit of Part 1 or Part 2, use the W output for the input of the register of Part 5 and connect the 8-bit output of the register to input A of the incrementer/decrementer logic block. You should now have a 4-mode counter as specified in Part 1. Simulate this circuit and test it for the four modes of operation. Start with model 11 forcing 8'b0 into the counter through the B input of the logic block.

Observe the waveforms and explain if it works as expected. Use pre-synthesis and post-synthesis descriptions in separate formations of the counter circuit.

7. Use two D-latches of Part 3 to build a master-slave D-type flip-flop. Build an 8-bit register using this flip-flop in a **generate** statement. Generate a synchronous resetting mechanism to enable the register to start with an initial value. Using a test bench, simulate this register and verify its clocking, and resetting operations. Write an 8-bit register using an **always** statement and annotate the delay values from the structural description into this description.
8. Take the circuit of Part 1 or Part 2, use the W output for the input of the register of Part 5 and connect the 8-bit output of the register of Part 7 to input A of the incrementer/decrementer logic block. Use the B input of the logic block as the parallel input of the counter. You should now have a 4-mode counter as specified in Part 1. Simulate this circuit and test it for the four modes of operation. Start with model 11 forcing 8'b0 into the counter through the B input of the logic block. Observe the waveforms and explain if it works as expected. Use pre-synthesis and post-synthesis descriptions in separate formations of the counter circuit.
9. Cascade two units of the module of Part 8 to form a 16-bit counter.

Deliverables:

Generate a report that includes items discussed below for each parts of this CA. Do not show any code without first showing its circuit diagram.

- A. Show the circuit diagram that you are analyzing. Show gates and/or transistors according to the specified delays.
- B. Generate a screen-shot of your Yosys synthesis results.
- C. Hand-simulate or analyze the circuits that you use in SystemVerilog testbenches. For example, indicate what values you expect for the worst-case delays and express your reason for that.
- D. Show your SystemVerilog descriptions of the design you are simulating and the testbench for it. Best is to use the Snip tools to get the image from Notepad++. This way you see all keywords and indentations. Make sure your SV codes are properly indented and all line-up rules are followed.
- E. Show an image of the project that you have created in ModelSim for the simulation of your circuit.
- F. When simulations are complete, or even if you have a partial simulation, include an image of the output waveforms showing signal names being displayed.

Make a PDF file of your report and name it with the format shown below:

FirstinitialLastnameStudentnumber-CAn-ECEmmm

Where *nn* is a two-digit number for the Computer Assignment, *mmm* is the three-digit course number under which you are registered, and hopefully you know the rest. For the *Firstinitial* use only one character. For *Lastname* and for the multi-part last names use the part you are most identified with. Use the last five digits of your student id (exclude 8101) for the *Studentnumber* field of the report file name.